

FYS5419/9419 May 6, 2026



QNN

$$|\psi(x, \epsilon)\rangle = U(\epsilon) U(x) |0\rangle^{\otimes n}$$

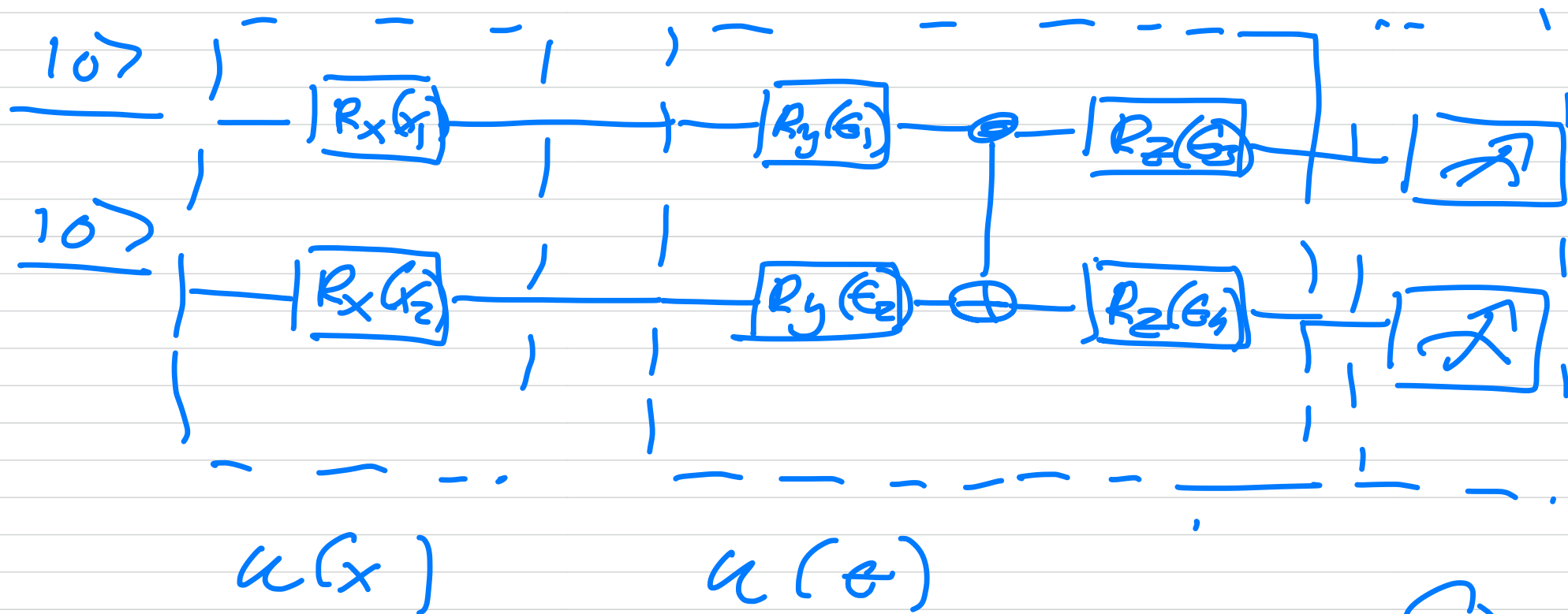
— variational ansatz

$$\epsilon = \{\epsilon_1, \epsilon_2, \dots, \epsilon_p\}$$

$$\epsilon_i \in \mathbb{R}$$

— feature map (data encoding)

Two qubits



variational

part

$$U_{\theta} = \prod_e e^{-i \epsilon_e P_e / 2} \quad P_e \in \{x, y, z\}^{\otimes n}$$

Optimization part

$$\text{Loss/cost} = \sum_k (f(x_k, \theta) - y_k)^2$$

input / (x_k, y_k)

output

parameter-shift rule

$$\partial_{\theta_k} f = \frac{1}{2} \left[f(\theta_k + \frac{\pi}{2}) - f(\theta_k - \frac{\pi}{2}) \right]$$

Gradient descent

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\uparrow f}$$

Natural gradient / stochastic reconfiguration

$$\Theta \leftarrow \Theta - \gamma F^{-1} \nabla_{\Theta} L$$

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Fisher info

F is a matrix which approximates the full Hessian matrix

VQE parallel
variational wave function

$$|\psi(\theta)\rangle \quad \theta = \{\theta_1, \theta_2, \dots, \theta_p\}$$

$$|\bar{\psi}(\theta)\rangle = \frac{|\psi(\theta)\rangle}{\sqrt{\langle \psi(\theta) | \psi(\theta) \rangle}}$$

$$E(\theta) = \frac{\langle \psi(\theta) | \mathcal{H} | \psi(\theta) \rangle}{\langle \psi(\theta) | \psi(\theta) \rangle}$$

Technicality: exact
imaginary-time evolution
 $\tau = it$

$$|\psi(\tau)\rangle = e^{-\tau \mathcal{H}} |\psi(0)\rangle$$

$$\mathcal{H}|n\rangle = E_n |n\rangle$$

$$|\psi(0)\rangle = \sum_n c_n |n\rangle$$

$$|\psi(\tau)\rangle = \sum_n c_n e^{-\tau E_n} |n\rangle$$

if $c_0 \neq 0$ for large τ ($E_0 < E_1 < E_2 \dots$)

$$|4(\gamma)\rangle \propto e^{-\gamma E_0} |0\rangle$$

idea of SR (Natural gradients)

Small time step $\delta\tau$

$$e^{-\delta\tau \mathcal{H}} = 1 - \delta\tau \mathcal{H} + \mathcal{O}(\delta\tau^2)$$

Exact propagator state

$$|4'_{\text{exact}}\rangle \simeq (1 - \delta\tau \mathcal{H}) |4(\epsilon)\rangle$$

First-order norm-preserving form

$$|\psi'_{\text{exact}}\rangle \approx [1 - \delta^2 (H - E(\psi))] |\psi\rangle$$

why subtract $E(\psi)$

$$E = \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle}$$

$$\langle \psi | (H - E) | \psi \rangle = 0$$

The vector $(H - E) | \psi \rangle$
is orthogonal to $|\psi\rangle$

$$|\psi(e + \delta e)\rangle =$$

$$|\psi(e)\rangle + \sum_k \delta e_k \frac{\partial}{\partial e_k} |\psi(e)\rangle + o(\delta e^2)$$

Define tangent vector

$$|\psi_k\rangle = \frac{\partial}{\partial e_k} |\psi(e)\rangle$$

Compare (First order)

$$|\psi'_{\text{var}}\rangle = |\psi(\vec{\theta})\rangle + \sum_k c_k |\phi_k\rangle$$

$$|\psi'_{\text{exact}}\rangle = [1 - \delta^2 (H - E)] |\psi\rangle$$

$$|\Delta\rangle = |\psi'_{\text{var}}\rangle - |\psi'_{\text{exact}}\rangle$$

Minimize

$$\mathcal{D} = \langle \Delta | \Delta \rangle$$

$$D = \sum_{i,j} \delta E_i \delta E_j \langle \psi_i | \psi_j \rangle$$

(to first order in δ^2)

$$+ 2\delta^2 \sum_i \delta E_i \langle \psi_i | (H - E) | \psi_i \rangle$$

optimize with respect to δE_i

$$\frac{\partial D}{\partial (\delta E_i)} = 2 \sum_j \langle \psi_i | \psi_j \rangle \delta E_j + 2\delta^2 \text{Re} \langle \psi_i | (H - E) | \psi_i \rangle = 0$$

$\Rightarrow$

$$\sum_j \langle \psi_i | \psi_j \rangle \delta E_j =$$

$$- \delta^2 \operatorname{Re} \langle \psi_i | H - E(\psi) | \psi_i \rangle$$

projected imaginary-time
eg. in the tangent

basis $\{ |\psi_i\rangle \}$

$$|\psi(E)\rangle = |\psi\rangle$$

$$|\bar{\psi}\rangle = \frac{|\psi\rangle}{\sqrt{\langle\psi|\psi\rangle}}$$

normalized tangent vectors

$$|\bar{\psi}_k\rangle = \frac{\partial}{\partial E_k} |\bar{\psi}\rangle$$

$$\sum_j \langle \bar{\psi}_k | \bar{\psi}_j \rangle \delta E_j = -\delta^2 \times \langle \bar{\psi}_k | H - E | \psi \rangle$$

Digression: log-derivative

$$O_K(x) = \frac{1}{\psi(x)} \frac{\partial \psi(x)}{\partial \theta_K} = \frac{\partial \ln \psi(x)}{\partial \theta_K}$$

For VQE the probability
is

$$p(x) = \frac{|\psi(x)|^2}{\langle \psi | \psi \rangle}$$

$$\langle A \rangle = \sum_x p(x) A(x)$$

$$\langle \bar{\psi}_i | \bar{\psi}_j \rangle = \langle O_i^* O_j \rangle - \langle O_i^* \rangle \langle O_j \rangle$$

Real wave functions and parameters

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle$$

S_{ij} is the covariance matrix of the log-derivatives

$$\langle \bar{\psi}_1 | H - E | \bar{\psi} \rangle$$

Define $E_L(x) = \frac{1}{\bar{\psi}(x)} H \bar{\psi}(x)$

$$g_1' = \text{Re} [\langle \bar{\psi}_1 (H - E) | \bar{\psi} \rangle]$$

$$= \langle O_1' E_L \rangle - \langle O_1' \rangle \langle E_L \rangle$$

$$= \frac{\partial E_L}{\partial G_1'} \cdot \frac{1}{2}$$

Putting together

$$\sum S_{ij} \delta E_j = -\delta \mathcal{F} \cdot g_i$$

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$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle$$

$$g_i = \langle O_i E \rangle - \langle O_i \rangle \langle E \rangle$$

For energy

$$G_D(\text{standard}) + \text{variants (adams)}$$
$$E \leftarrow E - \mu g$$

SR / normal gradient

$$\vec{E} \leftarrow \vec{E} - \underbrace{\delta \tilde{y}^T \tilde{S}^{-1}}_{\text{depends on}} \vec{g}$$

depends on

$$\frac{\partial \bar{\psi}(\theta)}{\partial \theta_k}$$

- reduced sensitivity to specific parametrizations
- better conditioning
- close relation $\psi(\theta)$

connection with natural
gradient

Squared distance between
neural network states

$$ds^2 = \sum_{i,j} S_{ij} d\theta_i d\theta_j$$

steepest descent in this
metric

$$\vec{\delta\theta} = -S^{-1} \cdot \vec{g}$$